

SE4095 — Social Network: Theory and Practice

PART 2 OF 3

Network Measures

Centrality, friendship patterns, and similarity — with every formula worked through numerically.

Covers: degree, eigenvector, Katz and PageRank centrality · betweenness and closeness centrality · comparing all six measures and knowing which to use · transitivity and reciprocity · social balance and status theory · structural equivalence, Jaccard and cosine similarity.

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Topic 3 — Centrality: Who Matters in a Network?

Why do we need measures at all?

Part 1 gave you the vocabulary (nodes, edges, paths). Now you need *numbers* so you can rank nodes and compare networks. The slides frame this around three questions: **Who are the central/influential individuals?** (centrality — this topic), **what interaction patterns are common between friends?** (reciprocity, transitivity, balance, status — Topic 4), and **who are the like-minded users?** (similarity — Topic 5).

Centrality itself splits into two fundamentally different questions, and mixing them up is the single most common source of lost exam marks in this topic:

Question being asked	What you're measuring	Measures that answer it
"How important is this node because of who it's connected to? "	Importance via connections (popularity / prestige)	Degree, Eigenvector, Katz, PageRank
"How important is this node because of how it connects others? "	Importance as a broker / information pathway	Betweenness, Closeness

Degree Centrality

Degree Centrality

WHAT IS IT?

FORMULA

$$C_d(v_i) = d_i$$

d_i the degree of node v_i — its number of length-1 paths (direct connections).

In directed graphs, split into in-degree and out-degree (in practice, **in-degree is used more often** — it captures who *receives* attention, which is usually what "important" means).

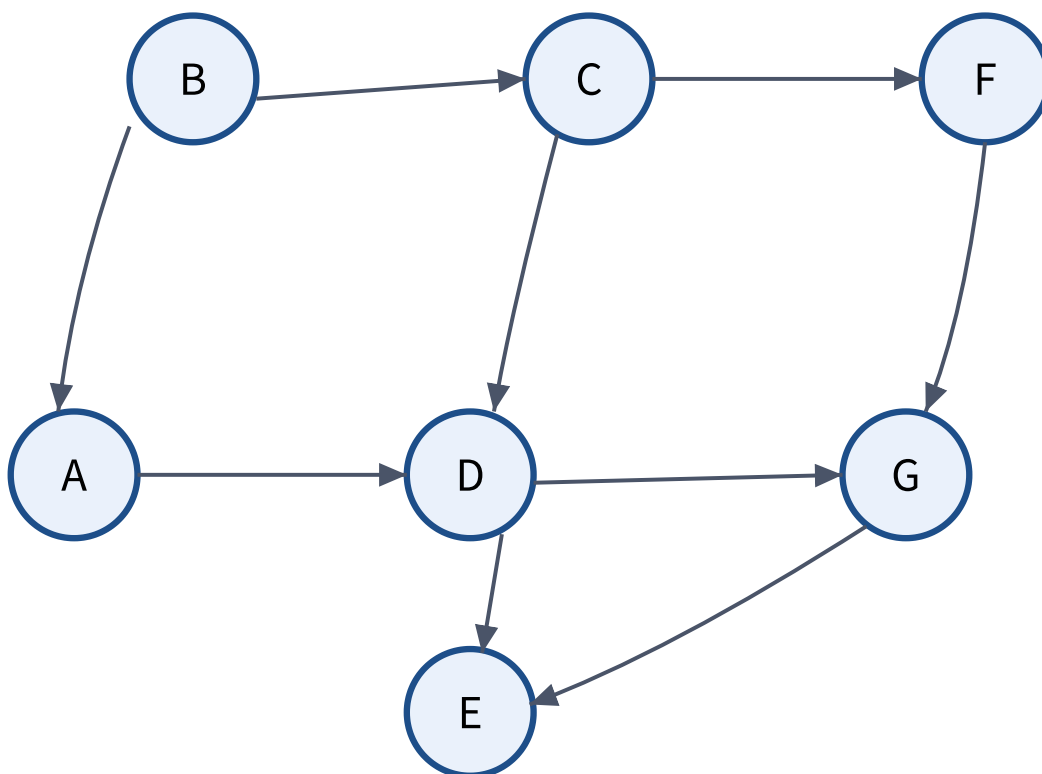
NORMALIZATION — 3 ways, pick based on what you're comparing

- By max possible degree: $C_d^{norm}(v_i) = \frac{d_i}{n-1}$ — lets you compare nodes *across networks of different sizes*.
- By the maximum degree actually observed in this network.
- By the sum of all degrees (turns it into a fraction of "total attention" in the network).

WORKED EXAMPLE – Degree centrality, directed graph, normalized by $n-1$

Node	In-degree	Out-degree	Centrality ($d_i/6$)	Rank
A	1	3	$1/2$	1
B	1	2	$1/3$	3
C	2	3	$1/2$	1
D	3	1	$1/6$	5
E	2	1	$1/6$	5
F	2	2	$1/3$	3
G	2	1	$1/6$	5

Check the convention: here "Centrality" = out-degree $\div$ ($n-1$) = out-degree/6 (e.g. A: out-degree 3 $\rightarrow$ $3/6 = 1/2$; D: out-degree 1 $\rightarrow$ $1/6$) — *not* total (in+out) degree. Always confirm which convention a question is using before you compute; in/out/total all give different rankings on a directed graph.



The A–G directed graph used across the degree-centrality, and PageRank worked examples below — the same graph is reused deliberately so you can see how each measure ranks the same nodes differently.

What you want to know	Use
How many people does this person follow?	Out-degree
How many people follow this person?	In-degree
Overall connectivity in an undirected network	Degree
Who receives information from many people?	In-degree
Who can send information to many people?	Out-degree

Eigenvector Centrality

Eigenvector Centrality

WHAT PROBLEM DOES IT SOLVE?

Degree centrality treats every connection as equally valuable. But intuitively, being friends with 3 highly influential people should count for more than being friends with 3 nobodies. Having more *important* friends is a stronger signal than just having more friends. Eigenvector centrality (Phillip Bonacich) generalizes degree centrality by weighting each neighbor's contribution by *that neighbor's own centrality*.

HOW DOES IT WORK?

We want $C_e(v_i)$ to be higher when important neighbors point to v_i . Assuming v_i 's centrality is the sum of its neighbors' centralities (with a normalizing constant), this leads to a self-referential equation:

FORMULA — Eigenvector centrality

$$\lambda \mathbf{C}_e = \mathbf{A}^T \mathbf{C}_e$$

This says the centrality vector $\mathbf{C}_e$ must be an eigenvector of the adjacency matrix $\mathbf{A}^T$ (or $\mathbf{A}$ itself, for undirected graphs), with λ the corresponding eigenvalue. A matrix has many eigenvalue/eigenvector pairs — we always take the one for the **largest** eigenvalue, because that's the only one guaranteed to give all-positive, meaningful centrality values (Perron–Frobenius).

WORKED EXAMPLE 1 – Eigenvector centrality on a 3-node path v1-v2-v3

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Solving $\det(A - \lambda I) = 0$ gives $\lambda(2 - \lambda^2) = 0$, so $\lambda \in \{0, -\sqrt{2}, +\sqrt{2}\}$. Take the largest, $\lambda = \sqrt{2}$. Solving $(A - \sqrt{2} I)C_e = 0$ and normalizing to unit length gives:

$$C_e = \begin{bmatrix} 1/2 \\ \sqrt{2}/2 \\ 1/2 \end{bmatrix} \approx \begin{bmatrix} 0.50 \\ 0.71 \\ 0.50 \end{bmatrix}$$

Interpretation: v2 (the middle node, connected to both others) gets the highest score — exactly matching intuition, since it's the only node adjacent to two neighbors instead of one.

WORKED EXAMPLE 2 – 5-node graph

For a 5-node graph (v1–v5) with a denser adjacency matrix, the eigenvalues come out to (2.68, -1.74, -1.27, 0.33, 0.00). Taking $\lambda_{\max} = 2.68$ and solving for its eigenvector gives:

$$C_e = \begin{bmatrix} 0.4119 \\ 0.5825 \\ 0.4119 \\ 0.5237 \\ 0.2169 \end{bmatrix}$$

Node v2 (index 2) has the highest score — check the adjacency matrix and you'll find it's the node with the most connections *and* the best-connected neighbors, both.

Situation	Use
"Who has the most friends?"	Degree centrality
"Who is connected to other important people?" / "popular because connected to popular people"	Eigenvector centrality
Simple measure of direct connectivity only	Degree centrality
Importance based on the whole network's structure, not just direct ties	Eigenvector centrality

Katz Centrality

Katz Centrality

WHAT PROBLEM DOES IT SOLVE?

Eigenvector centrality has a specific failure mode on directed graphs: centrality only passes over *outgoing* edges, so a node in a directed acyclic graph (DAG) — or any node with no incoming edges from already-central nodes — can end up with centrality exactly **zero**, even if it has plenty of connections. Katz fixes this by adding a small constant **bias term β** that every node gets regardless of its neighbors, so no one is stuck at zero.

HOW DOES IT WORK?

FORMULA — Katz centrality

$$\mathbf{C}_{Katz} = \beta(\mathbf{I} - \alpha\mathbf{A}^T)^{-1} \cdot \mathbf{1}$$

α controlling term — how much weight to give neighbors' centrality.

β bias term — the flat baseline every node gets.

$\mathbf{1}$ the all-ones vector.

When $\alpha=0$, every node gets the same score β (pure baseline, eigenvector effect switched off entirely). As α grows, the network-structure effect dominates over the flat baseline. For $(\mathbf{I}-\alpha\mathbf{A}^T)$ to be invertible, α must be chosen so $\alpha < 1/\lambda$, where λ is the largest eigenvalue of $\mathbf{A}^T$ (found efficiently via the power method).

WORKED EXAMPLE — Katz centrality

5-node graph, eigenvalues $-1.68, -1.0, -1.0, 0.35, 3.32$. We need $\alpha < 1/3.32 \approx 0.301$, so take $\alpha=0.25$ and $\beta=0.2$:

$$\mathbf{C}_{Katz} = \begin{bmatrix} 1.14 \\ 1.31 \\ 1.31 \\ 1.14 \\ 0.85 \end{bmatrix}$$

Nodes 2 and 3 (both scoring 1.31) come out most important — and critically, **no node scored zero**, which is exactly the property Katz was built to guarantee.

EXAM TIP — common mistake

Don't confuse α and β . α controls how much neighbors' importance matters (the "eigenvector-like" part); β is the flat floor everyone gets regardless of position. If a question asks "how does Katz centrality avoid zero-centrality nodes," the answer is β , not α .

PageRank

PageRank

WHAT PROBLEM DOES IT SOLVE?

Katz has its own failure mode: once a node becomes an authority (high centrality), it passes its *entire* centrality value along **every one** of its outgoing edges. That's unrealistic — not everyone a celebrity follows is themselves famous. PageRank's fix: a node divides its centrality *among* its outgoing links, so each neighbor gets only a fraction of it (proportional to $1/\text{out-degree}$).

HOW DOES IT WORK?

FORMULA — PageRank (conceptual form)

Each node passes its centrality divided by its out-degree to every outgoing neighbor, plus the same α/β bias structure as Katz. For undirected graphs, the largest eigenvalue of $A^T D^{-1}$ is exactly 1, so the constraint simplifies to $\alpha < 1$.

What if out-degree is zero? (a "dangling node," e.g. a webpage with no outgoing links) — this needs special handling (commonly: redistribute its centrality evenly across all nodes) or the divide-by-zero breaks the formula.

WORKED EXAMPLE — PageRank via the power method, on the same A-G graph

With $\alpha=0.95$, $\beta=0.1$, start every node at $1/7$ and iteratively spread centrality along out-links divided by out-degree. Selected steps from the full 15-round trace:

Step	A	B	C	D	E	F	G
1 (start)	0.143	0.143	0.143	0.143	0.143	0.143	0.143
2	0.071	0.048	0.190	0.167	0.190	0.119	0.214
15 (converged)	0.039	0.078	0.232	0.150	0.162	0.118	0.220
Final rank	7	6	1	4	3	5	2

How to read this: the values in each row always sum to 1.000 — PageRank redistributes a fixed total amount of "importance" around the network rather than creating new importance. By round 15, the numbers have essentially stabilized (this is what "convergence" means) — C is the clear winner, not because it has the most raw connections, but because of *whose* links point to it.

EXAM TIP — the power method / Markov chain connection

The slides explicitly show PageRank computed two ways: repeated matrix multiplication (the "power method") and as a Markov chain random walk (recall Random Walks from Part 1 — a walker moves to a random neighbor at each step, weighted by out-degree). Both converge to the *same* answer, because PageRank's score for a node is literally "the long-run probability a random walker is standing on it." If your teacher asks you to justify PageRank intuitively, "it's where a random surfer ends up spending their time" is the correct one-line answer.

"WHICH CENTRALITY SHOULD I USE?" – the classic exam scenario

This exact wording appears in the slides: **"A network needs to identify the most influential nodes."** There's no single correct answer without more context — that's the trap. Ask: does "influential" mean *directly popular* (degree), *connected to other important people* (eigenvector), does the network have *directed edges where the celebrity-dilution problem matters* (PageRank over Katz), or does it need *robustness to zero-centrality dead ends* (Katz/PageRank over plain eigenvector)? A strong exam answer names the specific reason, not just the measure.

Betweenness Centrality**Betweenness Centrality****WHAT PROBLEM DOES IT SOLVE?**

All four measures above ask "how important is this node because of who it's connected to." Betweenness asks a completely different question: **how important is this node as a bridge connecting other people?** A node can have very few connections and still be critical, if those few connections are the only path between two otherwise-separate parts of the network.

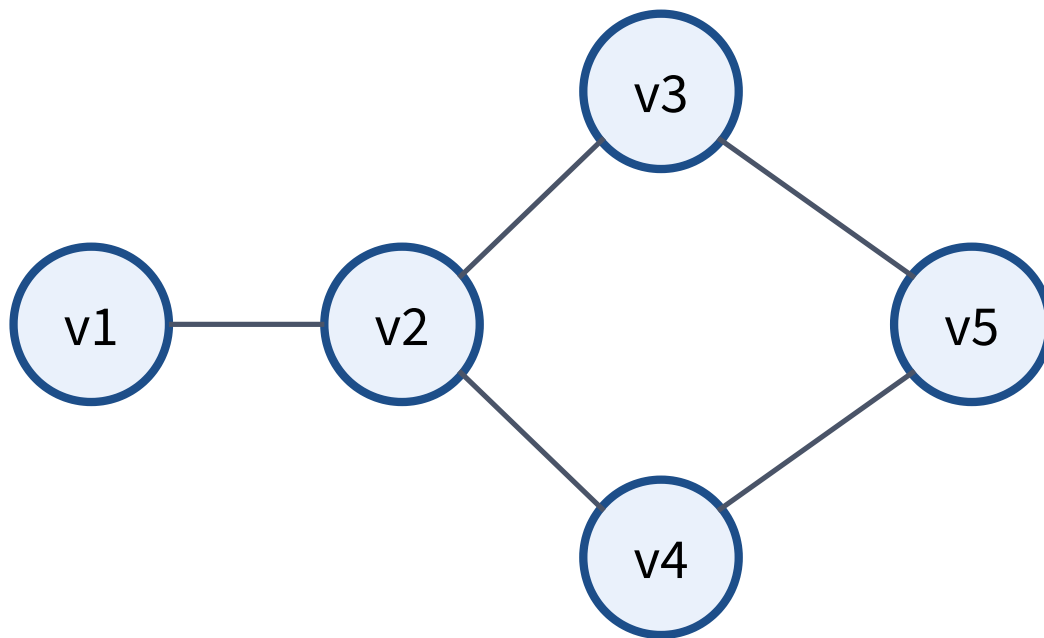
HOW DOES IT WORK?**FORMULA – Betweenness centrality**

$$C_b(v_i) = \sum_{s \neq v_i \neq t} \frac{\sigma_{st}(v_i)}{\sigma_{st}}$$

σ_{st} the total number of shortest paths from s to t .

$\sigma_{st}(v_i)$ how many of those shortest paths pass through v_i .

Normalized max value is $(n-1)(n-2)$ (or half that for undirected graphs, since each pair is counted once instead of twice).



The "diamond" graph used for both worked examples below — v2 and v5 are the two chokepoints; v3 and v4 are redundant alternatives of each other.

WORKED EXAMPLE — Betweenness on the diamond graph

Every shortest path between the two "arms" (v1 and v5) must pass through v2 or through {v3,v4} or through the far side. Counting every pair's shortest paths and who lies on them: $C_b(v2) = 2 \times 3.5 = 7$, $C_b(v3) = C_b(v4) = 2 \times 1.0 = 2$ each, $C_b(v5) = 2 \times 0.5 = 1$. (v1 scores 0 — it's an endpoint, never *between* two other nodes.)

Interpretation: v2 dominates betweenness even though its degree (3) isn't the network's highest, because it's the single chokepoint nearest the "spread out" side of the graph — exactly the broker role betweenness is designed to detect.

COMPUTATIONAL COST

The trivial approach (run Dijkstra from every node) costs $O(n^3)$. The standard method actually used is the **Brandes algorithm**: $O(nm)$ for unweighted graphs, $O(nm + n^2 \log n)$ for weighted graphs — a direct, memorable exam fact if asked about betweenness's computational cost.

Closeness Centrality

Closeness Centrality

WHAT IS IT?

The intuition: influential/central nodes can reach everyone else *quickly*. A node with a small average shortest-path distance to everyone else is "close" to the whole network.

FORMULA – Closeness centrality

$$C_c(v_i) = \frac{1}{\frac{1}{n-1} \sum_{j \neq i} l_{i,j}}$$

The reciprocal of the *average* shortest-path distance from v_i to every other node. Larger = closer to everyone = more central.

WORKED EXAMPLE – Closeness on the same diamond graph

Node	Distances to others	Average	C_c
v1	1,2,2,3	1.750	0.5 (lowest)
v2	1,1,1,2	1.250	0.8 (highest)
v3 / v4	2,1,2,2	1.750	0.66
v5	3,2,2,2	2.250	0.57

v2 wins again here — but notice betweenness and closeness agreed *on this particular graph*; they measure different things and often disagree on other graphs, which is exactly why the comparison below matters.

Comparing All Six Centralities

The Degree × Closeness × Betweenness Comparison

This is one of the highest-value tables in the entire course for scenario-recognition questions. When the three measures *agree* (all high or all low together), that's the unsurprising case. When they **disagree**, that disagreement itself tells you something structurally specific about the node's position.

	Low Betweenness	High Betweenness
Low Degree, Low Closeness	Node is embedded in a community that is far from the rest of the network	<i>Very rare</i> — a node with few ties that are nonetheless each critical bridges
Low Degree, High Closeness	Ego's connections are redundant — communication bypasses the node	Ego's few ties are crucial for network flow (a lean but critical broker)
High Degree, Low Closeness	Probably multiple paths in the network; ego is near many people, but so are many others	—
High Degree, High Closeness	Key node connected to important/active alters	Very rare — ego monopolizes ties from a small group to many others

EXAM TIP

Generally all three centralities are positively correlated (a well-connected node tends to be both close and between). When they're *not* correlated, that's specifically interesting and worth flagging in an answer — it's a common essay-style question to give you a node's three scores and ask "what does this combination tell you about its structural role?"

Measure	Main Purpose	Works Best When	Key Idea	Limitation	Exam Clue
Degree	Local popularity	Quick, cheap first pass	Count direct ties	Blind to the rest of the network	"most friends/followers"
Eigenvector	Importance via important neighbors	Undirected, well-connected networks	Weight ties by neighbor's own score	Can be exactly 0 for nodes with no path from central nodes (esp. directed/DAG)	"connected to other important people"
Katz	Fixes eigenvector's zero problem	Directed graphs, DAGs	Add a flat baseline β to every node	Over-values high out-degree nodes (celebrity problem)	"no node should be stuck at zero"
PageRank	Fixes Katz's celebrity-dilution problem	Web graphs, directed social graphs (who-follows-whom)	Split centrality across out-links	Needs special handling for dangling (zero out-degree) nodes	"not everyone a famous person follows is famous"
Betweenness	Brokerage / information control	Finding bottlenecks, critical connectors	Fraction of shortest paths through this node	Expensive to compute at scale (Brandes: $O(nm)$)	"connects two otherwise separate groups"
Closeness	Speed of reach	Fastest to spread info from this node	Inverse of average shortest-path distance	Sensitive to disconnected components (undefined/needs adjustment)	"reaches everyone else quickly"

Exam Checklist — Topic 3

- Two families: "important via connections" (degree, eigenvector, Katz, PageRank) vs "important as a broker" (betweenness, closeness).
- Degree: $C_d = d_i$; know all 3 normalizations and when each applies.
- Eigenvector: $\lambda C_e = A^T C_e$, always take the *largest* eigenvalue's eigenvector.
- Katz fixes eigenvector's "stuck at zero" problem via bias term β ; requires $\alpha < 1/\lambda_{\max}$.
- PageRank fixes Katz's "celebrity gives everyone full credit" problem by dividing centrality across out-links; equivalent to a random walk's long-run visit probability.
- Betweenness: fraction of shortest paths through a node; expensive (Brandes: $O(nm)$); measures brokerage, not popularity.
- Closeness: inverse of average shortest-path distance; measures speed of reach.
- The Degree×Closeness×Betweenness combination table — know what disagreement between them implies.
- **Common misconception:** higher degree does NOT imply higher betweenness or closeness — these can diverge sharply, and the divergence is often the exam's real question.

Quick Revision — Topic 3

Degree = count ties. Eigenvector = weight ties by neighbor importance (fails to zero on DAGs). Katz = eigenvector + flat baseline β (fixes the zero problem, but over-credits celebrities). PageRank = Katz, but divides centrality across out-links (fixes over-crediting) — equals a random walk's long-run location. Betweenness = fraction of shortest paths through a node (brokerage). Closeness = inverse average distance (speed of reach). All three of degree/closeness/betweenness usually agree; when they don't, that's structurally meaningful.

PRACTICE QUESTIONS — ANSWERED

Q1 (conceptual). Why can eigenvector centrality assign a score of exactly zero to a node that clearly has several connections?

A: In a directed graph (especially a DAG), eigenvector centrality only propagates importance backward along incoming edges from already-central nodes. If none of a node's in-neighbors have positive centrality (e.g. it only receives edges from other zero-centrality nodes, or sits in a part of the DAG with no path back from the "important" region), its score stays at zero regardless of its raw connection count.

Q2 (scenario). A researcher computes betweenness and closeness for every node in a company's email network and finds one employee with very low degree but the highest betweenness in the company. What does this most likely indicate, and why is this employee valuable despite few connections?

A: This matches the "Low Degree, High Betweenness" cell of the comparison table — the employee's few ties are individually crucial for network flow, likely acting as the sole bridge between two otherwise-separate teams or departments. They're valuable specifically as a broker, not as a popular hub — removing them would likely fragment communication between those groups.

Q3 (numerical). A node v has in-degree 4 and out-degree 1 in a 9-node directed graph. Using normalization by maximum possible degree, what is its degree centrality?

A: Using total degree (in+out) = 5, and $n-1 = 8$: $C_d^{\text{norm}}(v) = 5/8 = \mathbf{0.625}$. (If the question specifically wants in-degree only: $4/8 = 0.5$ — always check which convention the question specifies.)

Topic 4 — Friendship Patterns: Transitivity, Reciprocity, Balance & Status

Transitivity

Transitivity

WHAT IS IT?

DEFINITION

A relation R is transitive if aRc and cRb implies aRb . In social terms: **a friend of my friend is my friend.**

WHY DOES IT MATTER?

Transitivity in a social network pushes the graph toward being denser and closer to a complete graph — you can literally use the level of transitivity present as a way to measure how close a graph is to complete.

Reciprocity

Reciprocity

WHAT IS IT?

DEFINITION

Reciprocity is a simplified, 2-node version of transitivity — closed loops of length 2. "If you become my friend, I'll be yours." If v connects to u , and u connects back to v , that pair exhibits reciprocity.

FORMULA — Reciprocity

$$R = \frac{1}{m} \text{Tr}(A^2)$$

m = number of edges. $\text{Tr}(A^2)$ counts closed 2-step loops (i.e. mutual ties) across the whole adjacency matrix.

WORKED EXAMPLE – Reciprocity

3-node directed graph: $v1 \rightarrow v2$, $v2 \rightarrow v1$ (mutual), $v1 \rightarrow v3$, $v3 \rightarrow v2$ (one-way only).

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad m = 4$$

$$R = \frac{1}{4} \text{Tr}(A^2) = \frac{2}{4} = 0.5$$

Reciprocal nodes: $v1$ and $v2$ (they point to each other). $v3$'s tie to $v2$ is one-directional — it doesn't count.

Social Balance Theory

Social Balance Theory

WHAT PROBLEM DOES IT SOLVE?

People don't just have friends — they have friends *and* enemies, and some combinations of friend/enemy relationships feel psychologically "stable" while others feel tense and tend to resolve themselves over time (e.g. "the enemy of my enemy is my friend"). Balance theory formalizes exactly which triangles of relationships are stable.

HOW DOES IT WORK?

DEFINITION

Edges are signed: +1 for friendship, -1 for enmity. For a triangle of nodes i, j, k with edge signs w_{ij} , w_{jk} , w_{ik} , the triangle is **balanced** if and only if the product of its three edge signs is **positive**.

Edge signs (3 edges of the triangle)	Product	Balanced?	Real-world reading
+++	+	Balanced	All three mutual friends
+- -	+	Balanced	Two friends share a common enemy
- + -	+	Balanced	(same pattern, relabeled)
- - +	+	Balanced	(same pattern, relabeled)
++ -	-	Unbalanced	Two friends, one of them has a beef with the third — social tension, pressure to resolve
+ - +	-	Unbalanced	(same pattern, relabeled)
- + +	-	Unbalanced	(same pattern, relabeled)
- - -	-	Unbalanced	Three mutual enemies — surprisingly unstable; tends to resolve into one pair allying against the third

EXAM TIP — the shortcut

You never need to check all 8 combinations by hand: just multiply the three signs. Positive product = balanced. Negative product = unbalanced. This generalizes to *any* cycle in the graph, not just triangles: "for any cycle, if the multiplication of edge values is positive, the cycle is socially balanced."

Social Status Theory

Social Status Theory

WHAT IS IT?

Status theory is balance theory's directed cousin. Instead of friend/enemy, a directed '+' edge from X to Y means **Y has higher status than X**; a '-' edge means the reverse.

HOW DOES IT WORK?

Status is **consistent** when following the status arrows around a cycle gives you a coherent ranking (no contradictions). It's **inconsistent/unstable** when the arrows imply a contradiction — e.g. v1 says v2 is higher status, v2 says v3 is higher status, but v1 also says v3 is *lower* status than v1, directly contradicting the chain.

	Unstable configuration	Stable configuration
Edges	$v1 \rightarrow v2 (+), v2 \rightarrow v3 (+), v1 \rightarrow v3 (-)$	$v1 \rightarrow v2 (-), v2 \rightarrow v3 (-), v1 \rightarrow v3 (-)$
Reading	$v2 > v1$ and $v3 > v2$ implies $v3 > v1$, but the direct edge says $v3 < v1$ — contradiction	The direct comparison agrees with the chained comparison — no contradiction

"BALANCE OR STATUS?" — telling them apart

Both theories are about consistency in triangles, and questions often blur them together. The distinguishing test: **are the edges undirected friend/enemy relationships, or directed higher/lower comparisons?** Undirected signed ties (friend vs. foe) → balance theory. Directed ranking ties (who outranks whom) → status theory. If the scenario talks about alliances/rivalries, it's balance; if it talks about prestige/seniority/hierarchy, it's status.

Exam Checklist — Topic 4

- Transitivity: friend-of-friend-is-friend; pushes graphs toward completeness.
- Reciprocity: $R = (1/m)\text{Tr}(A^2)$; a 2-node special case of transitivity.
- Balance theory: multiply the 3 edge signs — positive = balanced. Learn the "enemy of my enemy" and "3 mutual enemies are unstable" readings.
- Status theory: directed version of balance; look for contradictions in a chained status comparison.
- **Common misconception:** "balanced" does not mean "all positive" — a triangle with two negative edges and one positive is also balanced (two friends united against a shared enemy).

Quick Revision — Topic 4

Transitivity = friend of a friend is a friend (drives density). Reciprocity = mutual ties, $R = \text{Tr}(A^2)/m$. Balance theory: multiply the triangle's 3 edge signs; positive = stable. Status theory: same idea but for directed higher/lower-rank edges — look for chained contradictions.

Topic 5 — Similarity: How Alike Are Two Nodes?

Structural Equivalence

WHAT PROBLEM DOES IT SOLVE?

You want to find "like-minded" users — but similarity here isn't about matching profile attributes, it's about matching *network position*. Two brothers share parents, grandparents, and often friends — that shared neighborhood is what makes them structurally similar, distinct from two random strangers who share nothing.

WHAT IS IT?

DEFINITION — Structural Equivalence

Two nodes are structurally equivalent to the degree that they share the same neighborhood — connect to the same set of other actors. The size of the shared neighborhood defines how similar two nodes are.

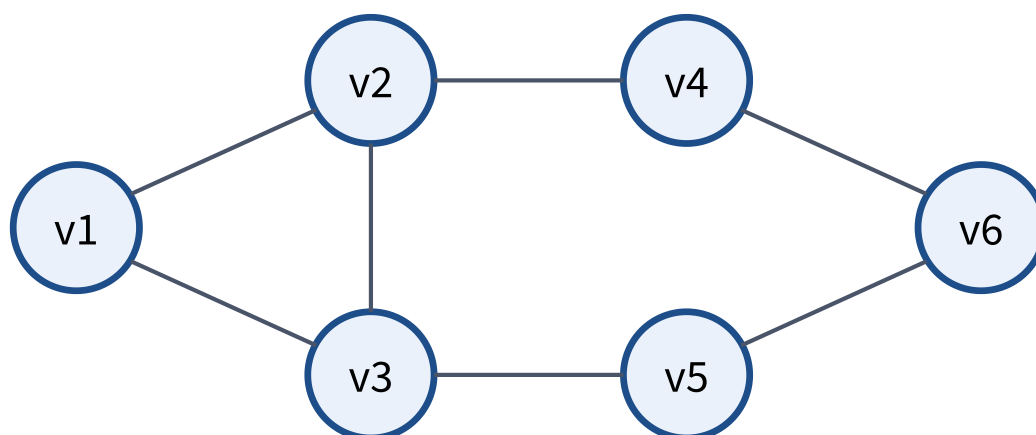
WHAT CAN GO WRONG, AND THE FIX?

If $N(v)$ excludes the node itself, two directly-connected nodes that happen to share no *other* common neighbor get assigned zero similarity — which is often wrong (they're connected to *each other*, after all). Fix: include each node in its own neighborhood before comparing.

FORMULA — Jaccard & Cosine similarity

$$\sigma_{Jaccard}(v_i, v_j) = \frac{|N(v_i) \cap N(v_j)|}{|N(v_i) \cup N(v_j)|} \quad \sigma_{Cosine}(v_i, v_j) = \frac{|N(v_i) \cap N(v_j)|}{\sqrt{|N(v_i)| |N(v_j)|}}$$

Jaccard: shared neighbors as a fraction of *all* neighbors either node has (a strict "how much overlap, relative to total spread" measure). Cosine: shared neighbors relative to the *geometric mean* of each node's neighborhood size (less harshly penalized when one node simply has many more neighbors than the other).



Similarity worked-example graph. $N(v2) = \{v1, v3, v4\}$, $N(v5) = \{v3, v6\}$.

WORKED EXAMPLE – Jaccard & Cosine similarity of v2 and v5

$N(v_2) = \{v_1, v_3, v_4\}$, $N(v_5) = \{v_3, v_6\}$. Intersection = $\{v_3\}$ (size 1). Union = $\{v_1, v_3, v_4, v_6\}$ (size 4).

$$\sigma_{Jaccard}(v_2, v_5) = \frac{1}{4} = 0.25 \quad \sigma_{Cosine}(v_2, v_5) = \frac{1}{\sqrt{3 \times 2}} = \frac{1}{\sqrt{6}} \approx 0.40$$

Why cosine is higher: cosine only "penalizes" for the product of neighborhood sizes, not their full union — so it's more forgiving when one node simply has a smaller neighborhood than the other. Jaccard is stricter because it counts every neighbor either node has, in the denominator.

Similarity Significance

Similarity Significance

WHAT PROBLEM DOES IT SOLVE?

A raw similarity score doesn't tell you if two nodes are *surprisingly* similar, or just similar by chance because they both happen to have many connections. Two very high-degree nodes will share *some* neighbors just by random luck — significance testing asks "is this overlap bigger than random chance alone would predict?"

FORMULA – Expected overlap under randomness

For nodes v_i, v_j with degrees d_i, d_j in a network of n nodes, the expected neighborhood overlap if neighbors were chosen at random is $\frac{d_i d_j}{n}$ — because v_i has a d_i/n chance of being any given node's neighbor, and v_j independently selects d_j neighbors.

SCENARIO RECOGNITION

"Are these two celebrities' fan bases meaningfully similar, or do they just both have millions of followers?" → similarity significance, not raw Jaccard/cosine. High-degree nodes need their overlap compared against the *expected* random overlap, or you'll systematically overrate how similar popular accounts are.

Exam Checklist — Topic 5

- Structural equivalence = shared neighborhood; include self in neighborhood to avoid false zeros between directly-connected nodes.
- Jaccard = intersection/union (strict); Cosine = intersection/geometric-mean-of-sizes (more forgiving of size mismatch).
- Similarity significance: expected random overlap = $d_i d_j / n$ — compare actual overlap against this before calling two nodes "meaningfully" similar.
- **Common misconception:** structural equivalence is about shared *connections*, not shared *attributes* (age, interests) — don't conflate it with homophily from Part 1, which is about why ties form, not how to measure position-similarity after the fact.

Quick Revision — Topic 5

Structural equivalence = do two nodes share the same neighbors. Jaccard = $\text{overlap}/\text{union}$ (strict). Cosine = $\text{overlap}/\sqrt{\text{geometric-mean}}$ (forgiving). Always sanity-check high-degree nodes' similarity against the random-chance expectation $d_i d_j / n$.

PRACTICE QUESTIONS — ANSWERED

Q1 (conceptual). Two nodes u and v are directly connected to each other but share zero other common neighbors. Using $N(v)$ defined to exclude v itself, what similarity score would they get, and why might this be misleading?

A: Zero — since $N(u)$ and $N(v)$ (excluding themselves) share no common elements, both Jaccard and cosine similarity would be 0. This is misleading because u and v are directly tied to each other, which is itself a strong similarity signal being thrown away — the standard fix is to include each node in its own neighborhood before comparing.

Q2 (scenario). A triangle of coworkers has relationships: Alice–Bob (friends, +), Bob–Carol (friends, +), Alice–Carol (rivals, –). Is this triangle balanced? What does balance theory predict will happen over time?

A: Product of signs = $(+)(+)(-) = \text{negative} \rightarrow$ **unbalanced**. Balance theory predicts social tension and pressure to resolve: either Alice and Carol eventually reconcile (all three become friends, giving +++, balanced), or Bob is forced to "pick a side" and one of his friendships sours, producing two enemies uniting against a shared friend-turned-target (a different balanced configuration).

Q3 (numerical). In a 100-node network, node X has degree 20 and node Y has degree 15. If X and Y actually share 8 common neighbors, is that more or less than expected by pure chance?

A: Expected overlap = $d_X d_Y / n = (20 \times 15) / 100 = 3$. The observed overlap of 8 is well above the random-chance expectation of 3 — so X and Y are *significantly* more similar than their degrees alone would predict, suggesting a real structural or topical reason they connect to the same people (not just a side-effect of both being well-connected).